

Abstract

The increasing adoption of information and communication technologies offers new opportunities to improve healthcare service delivery, yet many healthcare facilities still rely on manual appointment scheduling methods, leading to long waiting times, scheduling conflicts, inefficient record keeping, and limited communication between patients and healthcare providers. To address these challenges, this project proposes the design and future implementation of a web-based online doctor's appointment system that will streamline interactions among patients, doctors, and administrative staff. The system will allow patients to register online, manage their profiles, view available medical services, and schedule appointments at convenient times, while also supporting virtual consultations through Google Meet integration for remote access. It will operate using three main user roles: patient, doctor, and admin enabling doctors to manage their availability and appointments and administrators to oversee appointment approvals and system management. Developed with Next.js and TypeScript for the frontend, Node.js and Express.js for the backend, and PostgreSQL for secure data storage, the system will implement authentication via JSON Web Tokens and secure integration with Google services through OAuth 2.0. With a focus on usability, security, scalability, and performance, the system is expected to reduce administrative workload, enhance patient engagement, and improve communication between patients and healthcare providers, ultimately contributing to more efficient healthcare service delivery in the future.

1. Introduction

Healthcare services in many areas face challenges such as limited infrastructure, high patient demand, and reliance on manual appointment scheduling and record-keeping. These traditional methods often lead to long waiting times, scheduling conflicts, incomplete records, and poor communication between patients and healthcare providers. The lack of digital solutions further limits access and efficiency, particularly in remote or underserved areas.

This project proposes the design and future implementation of a web-based online doctor's appointment system. The system will allow patients to register online, manage profiles, view services, and schedule appointments. It will support virtual consultations through Google Meet and include role-based access for patients, doctors, and administrators. The system aims to reduce administrative workload, improve communication, and provide more efficient and accessible healthcare services.

1.1 Background

Healthcare systems in many developing countries face challenges such as limited infrastructure, high patient demand, and inefficient administrative processes. In Ethiopia, these challenges are particularly evident, as many healthcare facilities still rely on manual methods for managing patient appointments and medical records, including paper-based registration and phone calls. Such traditional approaches often lead to long waiting times, scheduling conflicts, incomplete or lost records, and poor communication between patients and healthcare providers. The limited use of digital technologies further restricts the accessibility and quality of healthcare, especially for patients in remote or underserved areas.

The growing need for efficient, timely, and well-organized healthcare services highlights the potential of digital solutions to address these challenges. Web-based appointment systems can streamline patient registration, scheduling, and record management while improving communication between patients and healthcare providers. By reducing administrative workload,

minimizing errors, and providing timely notifications, such systems can enhance patient satisfaction and overall service delivery, even in resource-limited settings.

This project proposes the design and future implementation of a web-based online doctor's appointment system tailored to the Ethiopian healthcare context. The system will enable patients to register online, manage their personal profiles, view available medical services, and schedule appointments efficiently. It will also support virtual consultations through Google Meet integration, improving access for patients who cannot easily visit healthcare facilities. By implementing role-based access for patients, doctors, and administrators, the system aims to enhance appointment management, streamline communication, and contribute to a more organized, accessible, and efficient healthcare service delivery in Ethiopia.

1.2 Statement of the Problem

Despite the critical role of healthcare services, many facilities continue to rely on manual and paper-based processes for patient registration, appointment scheduling, and record management. These traditional methods are often slow, inefficient, and prone to human error, resulting in long waiting times, missed or overlapping appointments, and incomplete patient records. Patients frequently face difficulties in accessing information about available services or doctors, while healthcare staff spend considerable time on administrative tasks that could otherwise be devoted to patient care.

Additionally, the lack of a centralized digital system makes it difficult to coordinate schedules, track patient history, and provide timely notifications or reminders for upcoming consultations. This gap not only affects the efficiency of healthcare providers but also reduces patient satisfaction and engagement, creating challenges for maintaining quality care. The growing demand for healthcare services further exacerbates these issues, making manual systems increasingly inadequate to meet the needs of both patients and providers.

Therefore, there is a pressing need for a reliable, user-friendly, and accessible digital system that can streamline appointment management, improve communication between patients and

healthcare staff, and reduce administrative workload. Such a system would allow patients to book and manage appointments conveniently, while enabling healthcare providers to monitor schedules, track patient history, and offer timely services more effectively. Implementing a web-based solution with virtual consultation capabilities and role-based access would address these challenges, improving overall healthcare service delivery and patient experience.

1.3 Objectives of the Project

1.3.1 General Objective

The general objective of this project is to design and implement a web-based online doctor's appointment system that improves the efficiency, accessibility, and quality of healthcare services by streamlining patient registration, appointment scheduling, and communication between patients and healthcare providers.

1.3.2 Specific Objectives

The specific objectives of the project include:

- To develop an online platform that allows patients to register, manage their profiles, and schedule appointments conveniently.
- To implement role-based access for patients, doctors, and administrators to ensure secure and organized management of appointments and patient data.
- To integrate virtual consultation support through Google Meet to facilitate remote healthcare services.
- To provide automated notifications and reminders for upcoming appointments and follow-ups.

- To display clear information about medical services and fees, helping patients make informed decisions.
- To reduce administrative workload for healthcare staff by providing an efficient digital system for managing appointments and patient records.

1.4 Purpose of the Project

The purpose of this project is to provide a digital solution that simplifies and improves the management of doctor appointments. By implementing a web-based system, patients will have a convenient way to register, schedule, and track their appointments, while healthcare providers will be able to manage schedules, access patient information, and communicate more effectively. The system also aims to support virtual consultations, reducing the need for physical visits and improving access to healthcare services. Overall, the project seeks to enhance the efficiency, accuracy, and quality of healthcare delivery, benefiting both patients and healthcare staff.

1.5 Scope of the Project

The scope of this project focuses on the development and future implementation of a web-based online doctor's appointment system. The main aspects included in the project are:

- **Patient Management**
 - Registration of new patients with personal, contact, and basic medical information.
 - Profile management, allowing patients to update their information and view appointment history.

- Online appointment scheduling with the ability to select services, doctors, dates, and times.
- Access to virtual consultation links through Google Meet for remote appointments.
- Receiving notifications and reminders for upcoming appointments and follow-ups.
- Providing feedback on services and consultations.

➤ **Doctor Management**

- Viewing scheduled appointments and patient information related to assigned consultations.
- Managing availability by setting working hours and available time slots for appointments.
- Accessing virtual consultation links for online sessions.
- Monitoring patient attendance and consultation history for better follow-up.

➤ **Administrator Management**

- Approving, rejecting, or rescheduling appointment requests.
- Managing patient and doctor records within the system.
- Overseeing system operation, ensuring smooth scheduling and communication.

- Managing notifications and reminders to keep patients and doctors informed.

➤ **System Features and Limitations**

- Support for role-based access control to ensure secure and organized operations.
- Display of service fees and supported payment methods without direct transaction processing.
- Integration with Google Meet for virtual appointments.
- Focused on appointment management; does not include treatment or direct payment processing modules.
- Designed with usability, security, scalability, and performance in mind to handle multiple concurrent users.

➤ **Operational Scope**

- The system will be accessible via web browsers on desktops and mobile devices.
- It will provide a centralized platform for managing appointments and patient-doctor interactions.
- The system will improve communication between patients, doctors, and administrative staff while reducing administrative workload.

1.6 Methodology

The methodology outlines the approach and techniques that will be used for the design, development, and testing of the web-based online doctor's appointment system. The methodology is divided into several sub-sections for clarity.

1.6.1 Data Gathering Techniques

- **Interviews:** Discussions with healthcare staff and administrators to understand current challenges and requirements.
- **Observation:** Reviewing current manual appointment systems to identify inefficiencies and workflow patterns.
- **Literature Review:** Studying existing online appointment systems and related research to incorporate best practices and modern technologies.

1.6.2 Design Methodology

- **Requirement Analysis:** Identifying functional and non-functional requirements for patients, doctors, and administrators.
- **System Modeling:** Developing use case diagrams, class diagrams, sequence diagrams, and activity diagrams to visualize system operations and interactions.
- **Interface Design:** Creating user-friendly interfaces to ensure accessibility for patients with varying levels of technical literacy.
- **Role-Based Access Design:** Defining access control for patients, doctors, and administrators to ensure secure and organized system operations.

1.6.3 Implementation Methodology

- **Frontend:** Next.js with TypeScript for a responsive, fast, and secure user interface.
- **Backend:** Node.js with Express.js for handling business logic, APIs, and Google Meet integration.
- **Database:** PostgreSQL for reliable and consistent storage of patient records, appointments, and user data.
- **Authentication & Security:** JSON Web Tokens (JWT) for secure login sessions, and OAuth 2.0 for integration with Google services.
- **System Documentation:** Comprehensive documentation will be maintained to guide users, administrators, and future developers.

1.6.4 Testing Methodology

- **Unit Testing:** Testing individual components of the system for correct functionality.
- **Integration Testing:** Ensuring that different modules (frontend, backend, database) work together as expected.
- **System Testing:** Conducted in two phases:
 - **Alpha Testing:** Internal testing by the development team to identify and fix initial bugs.
 - **Beta Testing:** Limited release to actual users (patients and staff) to gather feedback and identify issues in a real-world scenario.

1.6.5 Hardware and Software

- **Hardware:** Standard desktops or laptops with internet connectivity, capable of running modern web browsers.
- **Software:**
 - Frontend: Next.js (TypeScript)
 - Backend: Node.js, Express.js
 - Database: PostgreSQL
 - Authentication: JWT and OAuth 2.0
 - API: RESTful services

2. Requirement Analysis Description

2.1 Overview of the Existing System

Currently, most healthcare facilities rely on manual processes to manage patient appointments and records. Patients typically register in person at the reception desk, where their personal and medical information is recorded in physical logbooks or spreadsheets. Appointment schedules are often maintained manually, creating challenges in coordinating patient visits, avoiding conflicts, and tracking history.

Communication between patients and healthcare providers is mostly handled through phone calls or in-person inquiries, which can lead to missed information or delays. Doctors and

administrative staff spend considerable time managing these manual processes, which reduces the time available for direct patient care.

While this system has the advantage of being familiar to staff and low-cost, it is inefficient, error-prone, and difficult to scale, especially as patient volumes increase or when virtual consultations are needed. The lack of centralized digital management makes it difficult to track patient history, monitor appointments, and provide timely notifications or reminders.

2.1.1 Activities of the System

The current manual system for managing healthcare appointments involves several key activities performed by patients, doctors, and administrative staff. These activities include:

➤ **Patient Registration**

- Patients provide their personal and medical information at the reception.
- Administrative staff record this information manually in logbooks or spreadsheets.

➤ **Appointment Scheduling**

- Patients request appointments in person or via phone.
- Staff manually check the doctor's availability and assign time slots.
- Appointment details are written in physical appointment books or basic spreadsheets.

➤ **Consultation Management**

- Doctors review appointment schedules prepared by staff.

- Consultation notes and patient records are maintained on paper.
- Follow-ups are recorded manually, requiring staff to track patient visits.

➤ **Communication and Notifications**

- Appointment confirmations and reminders are given verbally or through phone calls.
- Patients may miss appointments due to lack of automated reminders.

➤ **Record Maintenance and Reporting**

- Staff are responsible for maintaining patient records, appointment history, and doctor schedules.
- Generating reports or monitoring trends is time-consuming and prone to errors due to the manual process.
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2.1.2 Problems of the Existing System

- Long waiting times due to manual registration and scheduling processes.
- Scheduling conflicts caused by the lack of a centralized system.
- Incomplete, lost, or damaged patient records because of paper-based documentation.
- Limited patient access to appointment information, making scheduling inconvenient.

- Communication gaps, as reminders and notifications are handled verbally or via phone calls.
- High administrative workload for staff managing appointments, records, and follow-ups.
- Inefficient reporting and monitoring due to reliance on manual record-keeping.

2.1.3 Weaknesses and Strengths

Weaknesses

- The manual appointment system is time-consuming, requiring staff to handle registration, scheduling, and record-keeping physically, which slows down the overall workflow.
- Scheduling conflicts are frequent because there is no centralized platform to coordinate patient appointments, often resulting in double bookings or missed appointments.
- Patient records are prone to loss, damage, or incompleteness due to reliance on paper-based documentation, making it difficult to maintain accurate medical history.
- Communication between patients and healthcare staff is limited and inefficient, as reminders, confirmations, and notifications are usually conveyed verbally or through phone calls.
- Administrative staff are burdened with repetitive tasks, reducing the time available for direct patient care and increasing the likelihood of human error.
- The system lacks scalability; as the number of patients increases, managing appointments and records becomes increasingly difficult and inefficient.

Strengths

- The current system is simple and familiar to staff, requiring minimal training to operate.
- It has a low initial cost since it does not rely on digital infrastructure or complex technology.
- Basic appointment management is achievable without depending on internet access or electronic devices.
- Staff have direct control over records and schedules, which can provide a sense of oversight in small facilities.

2.1.4 Business Rules

1. **Each patient must register with a unique identifier to ensure proper tracking of appointments and medical history.**

This ensures that every patient can be uniquely recognized by the system. It prevents confusion between patients with similar names and allows the system to maintain accurate records of appointments, consultations, and medical history. Unique identifiers also make it easier for administrators to manage data and generate reports.

2. **Appointments requested by patients must be approved or confirmed by the administrator before they are finalized.**

This rule ensures that appointments are properly scheduled and coordinated with the doctor's availability. Administrators can prevent double bookings or conflicts, ensuring that patients receive confirmed slots and that doctors' schedules remain organized. It also allows staff to intervene in case of errors or special requests.

3. **Administrators have full access to manage patient and doctor records, approve or reject appointments, and oversee overall system operations.**

Administrators act as the system managers, ensuring smooth operations. They can modify records, handle scheduling issues, and monitor system activity. This centralized control helps maintain accuracy, resolve conflicts, and provide support to both patients and doctors when needed.

4. **Virtual consultations are only accessible through automatically generated Google Meet links, which are provided to patients upon confirmation of their appointment.**

To ensure security and convenience, virtual consultation links are generated automatically by the system and sent to patients after their appointment is confirmed. This prevents unauthorized access and simplifies the process for patients and doctors, allowing them to join consultations without manual link sharing.

5. **Patients are responsible for providing accurate personal and contact information during registration to ensure proper scheduling and communication.**

Accurate patient information is critical for effective appointment management. Correct contact details allow the system to send notifications, reminders, and virtual consultation links. Errors or omissions can lead to missed appointments, confusion, or communication breakdowns.

6. **All appointments, records, and notifications must be stored securely to protect patient privacy and comply with data protection standards.**

Sensitive data must be encrypted and stored safely to prevent unauthorized access. Secure storage ensures compliance with privacy regulations, protects patient confidentiality, and maintains trust between patients, doctors, and administrators.

7. **System notifications and reminders must be sent automatically to patients and doctors to minimize missed appointments and improve communication.**

Automated notifications reduce human error and ensure that all parties are informed about upcoming appointments. Reminders help patients attend consultations on time and

allow doctors to prepare for scheduled visits, improving the overall efficiency of the healthcare service.

2.2 Overview of the Proposed System

The proposed system is a web-based online doctor's appointment platform designed to overcome the limitations of the existing manual system. It provides a centralized, digital solution for managing patient registration, appointment scheduling, and communication between patients, doctors, and administrators. The system aims to increase efficiency, reduce administrative workload, and improve patient experience.

➤ **Patient Registration and Profile Management**

Patients can create accounts with personal, contact, and basic medical information. This ensures that patient records are properly maintained, easy to access, and can be used to schedule appointments and track medical history accurately.

➤ **Role-Based Access Control**

The system distinguishes between patients, doctors, and administrators, giving each role specific permissions. This ensures security, prevents unauthorized access, and allows users to perform only the tasks relevant to their responsibilities.

➤ **Appointment Scheduling**

Patients can view available services, select doctors, and book appointments online. The system checks for scheduling conflicts and prevents double bookings, making the process faster, more accurate, and convenient.

➤ **Virtual Consultation Support (Google Meet Integration)**

Automatically generated Google Meet links are provided to patients and doctors for remote consultations. This allows patients to access medical services from home or

distant locations, improving accessibility and reducing the need for physical visits.

➤ **Automated Notifications and Reminders**

The system sends reminders to patients and doctors about upcoming appointments or follow-ups. This reduces missed appointments, keeps users informed, and improves overall communication.

➤ **Display of Service Fees and Payment Methods**

Patients can view the costs of services and available payment options. Although the system does not handle direct transactions, this feature helps patients plan their visits and reduces confusion about fees.

➤ **Admin Dashboard**

Administrators can approve or reject appointments, manage patient and doctor accounts, and monitor overall system activity. The dashboard provides centralized control to ensure smooth operation and maintain data accuracy.

➤ **Patient Portal and Feedback**

Patients can access their appointment history, view records, and provide feedback on services. This feature promotes patient engagement and allows healthcare providers to improve service quality.

2.2.2 Non-Functional Requirements

➤ **Usability**

The system must have a simple, intuitive, and user-friendly interface that can be easily used by patients with varying levels of technical literacy. Clear navigation, understandable instructions, and a responsive design are essential to encourage adoption and minimize errors during usage.

➤ **Security**

Patient data and appointment records must be protected from unauthorized access. The system will implement password encryption, role-based access control, and secure session management. Additionally, sensitive information such as personal and medical data must comply with privacy standards to ensure confidentiality and trust.

➤ **Reliability and Availability**

The system must be available 24/7 with minimal downtime. It should provide consistent performance under normal and peak usage, ensuring that patients can schedule appointments and access information at any time. Backup and recovery mechanisms should be in place to prevent data loss.

➤ **Scalability**

The system should be capable of handling multiple concurrent users, including patients, doctors, and administrators. It must efficiently manage growing amounts of data and increasing numbers of users without performance degradation.

➤ **Performance**

The system must respond quickly to user actions, such as logging in, scheduling appointments, or accessing virtual consultation links. Fast response times are critical for maintaining user satisfaction and preventing delays in healthcare service delivery.

➤ **Maintainability**

The system should be designed in a modular and organized way, allowing developers to update, fix, or enhance features without affecting the overall system. Proper documentation and clear code structure will support maintainability.

➤ **Portability**

The system must be accessible via different devices and web browsers, including desktops, laptops, and mobile devices. This ensures flexibility for patients and staff to use the system from various locations.

2.2.3 System Requirements (Hardware and Software)

The proposed online doctor's appointment system requires both hardware and software components to function effectively and provide a seamless experience for patients, doctors, and administrators. On the hardware side, the system is designed to be practical and accessible, requiring only standard computers or laptops with internet access. Patients can access the system using personal computers, laptops, or mobile devices with modern web browsers. Healthcare staff and administrators will also use standard office computers to manage appointments, patient information, and system operations. The server, which hosts the application and database, should be capable of handling multiple simultaneous users and storing patient records securely, but no specialized or high-end hardware is necessary. This makes the system feasible for healthcare facilities of varying sizes, including small clinics with limited resources.

On the software side, the system will use a layered web application architecture to ensure performance, security, and scalability. The frontend will be developed using Next.js with TypeScript, which allows the creation of a responsive, interactive, and user-friendly interface. This ensures that patients can easily navigate the system, book appointments, and access virtual consultations, while doctors and administrators can efficiently manage schedules and records. The backend will be implemented using Node.js with Express.js, providing a fast and efficient platform to handle application logic, API requests, and integration with external services such as Google Meet. PostgreSQL will serve as the relational database management system, ensuring reliable storage of patient records, appointment schedules, and system logs with high data integrity.

Security is a key requirement of the system, and it will be ensured using JSON Web Tokens (JWT) for secure authentication, session management, and role-based access control. OAuth 2.0 will be used for integrating Google Meet for virtual consultations, ensuring that only authorized users can access online appointments. RESTful APIs will facilitate smooth communication between the frontend and backend, enabling modular development and easy maintenance. The software stack has been chosen to ensure the system is scalable, secure, and capable of handling future enhancements without requiring major changes.

2.3 Constraints and Assumptions

The development and operation of the proposed online doctor's appointment system are subject to several constraints and assumptions that define its scope and ensure realistic expectations. One of the main constraints is that the system will not handle direct online payment processing. While it can display service fees and available payment methods, all transactions must be completed outside the system, which simplifies development and reduces potential security risks. Another constraint is that virtual consultations rely on stable internet connectivity; both patients and doctors must have access to a reliable network to participate in online appointments. Additionally, the system is focused on appointment management, scheduling, and record-keeping, and does not include modules for diagnosing, prescribing medication, or other advanced medical functionalities.

Several assumptions are made regarding system users and operations. It is assumed that patients have basic familiarity with using web browsers or mobile devices, allowing them to navigate the platform, register, and schedule appointments independently. Doctors and administrators are expected to maintain accurate schedules, update availability, and ensure that patient information is entered correctly. It is also assumed that patients will provide accurate personal and contact details during registration, enabling proper scheduling, notifications, and virtual consultation access. Furthermore, the system assumes that the healthcare facility provides adequate infrastructure, such as computers and internet access, to support the platform's operations.

These constraints and assumptions provide a framework for the development and deployment of the system, ensuring that it operates effectively within practical limitations while meeting the needs of patients, doctors, and administrators. By clearly defining these boundaries, the system can be designed, implemented, and maintained in a manner that is both efficient and sustainable.

3. System Modeling

System modeling provides a visual representation of how the online doctor's appointment system operates. It helps in understanding interactions between users and the system, clarifying functional requirements, and guiding the design and implementation phases.

3.1 Use Case Model

The use case model represents the interactions between the system and its users (actors). It highlights the main functions of the system and shows how patients, doctors, and administrators interact with it.

3.1.1 Actor Specification

- **Patient:** Registers, schedules appointments, accesses virtual consultations, and provides feedback.
- **Doctor:** Manages availability, views scheduled appointments, and conducts consultations.
- **Administrator:** Approves or rejects appointments, manages patient and doctor records, and monitors system activity.

3.1.2 Use Case Diagram

- Patient: Register, schedule appointment, access virtual consultation, receive notifications, provide feedback.
- Doctor: Manage availability, view appointments, conduct consultations.

- Administrator: Approve/reject appointments, manage accounts, monitor system operations.

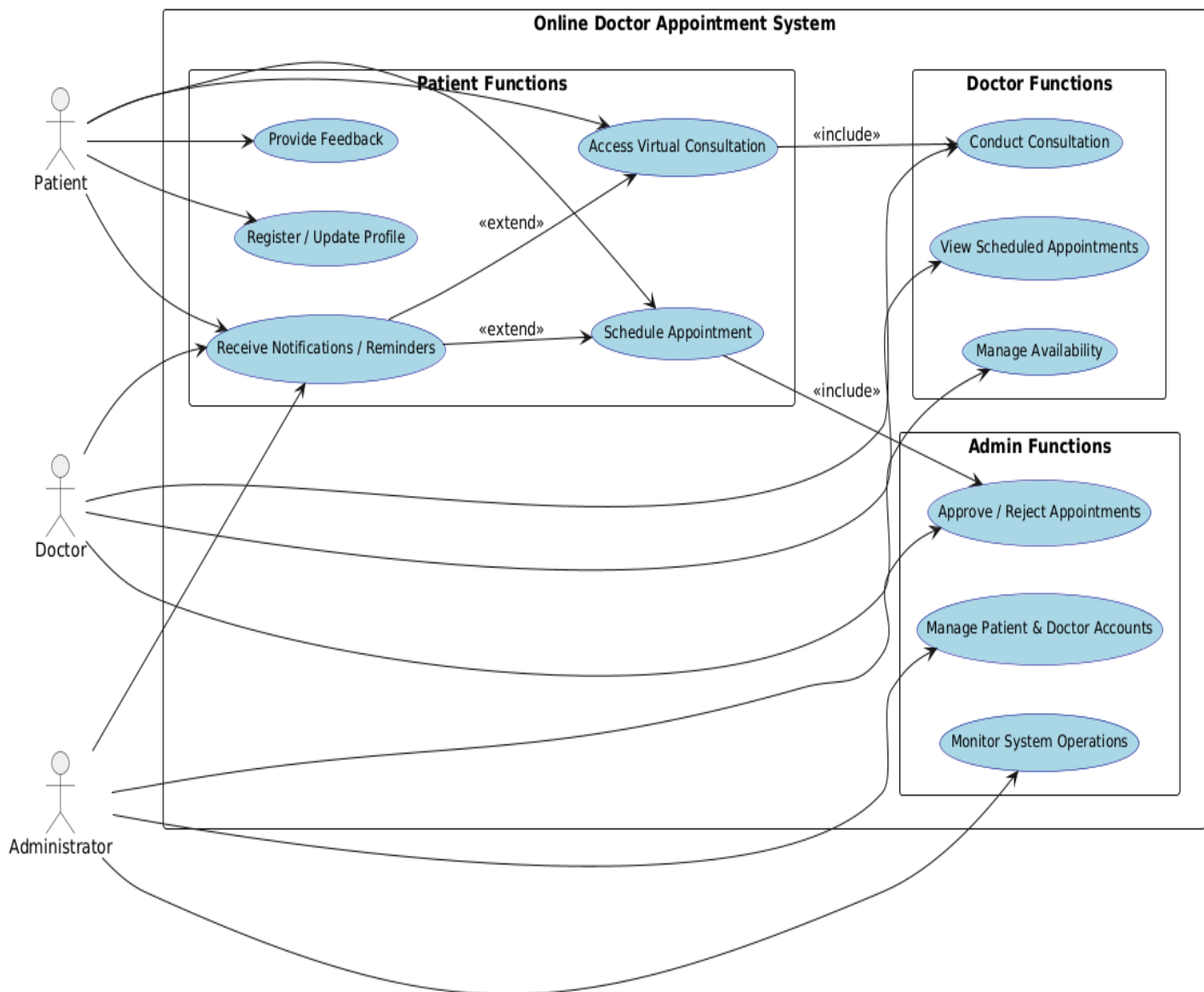


Figure 1 overall use-case diagram

Patient Use Case Diagram

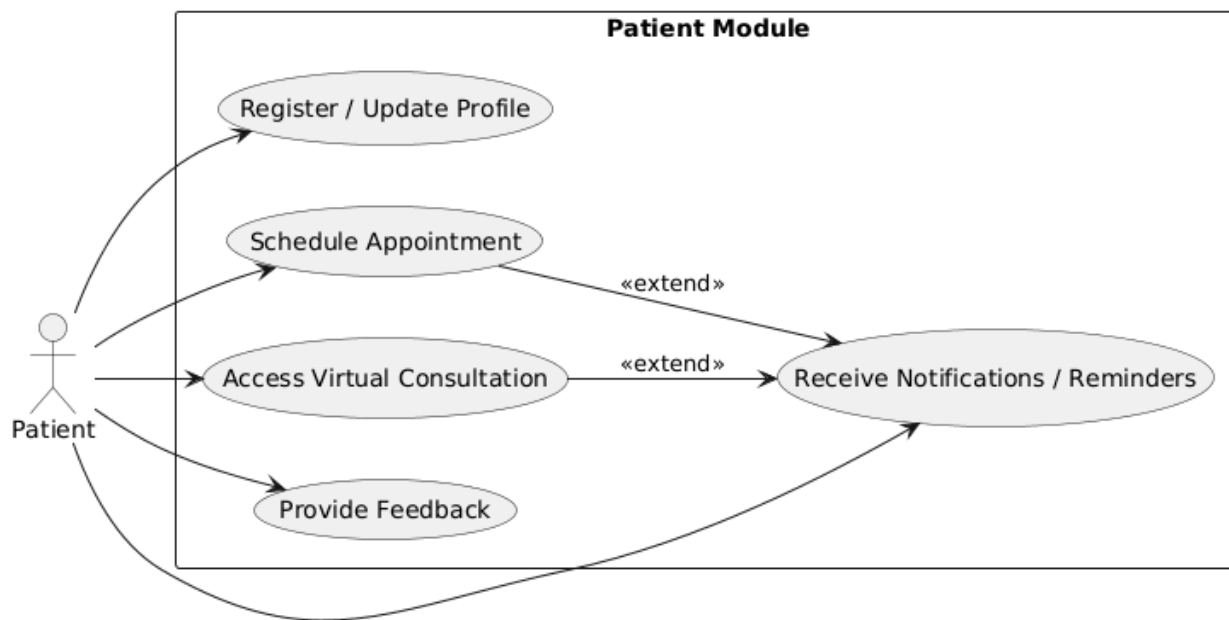


Figure 2 Patient Use Case Diagram

Doctor Use Case Diagram

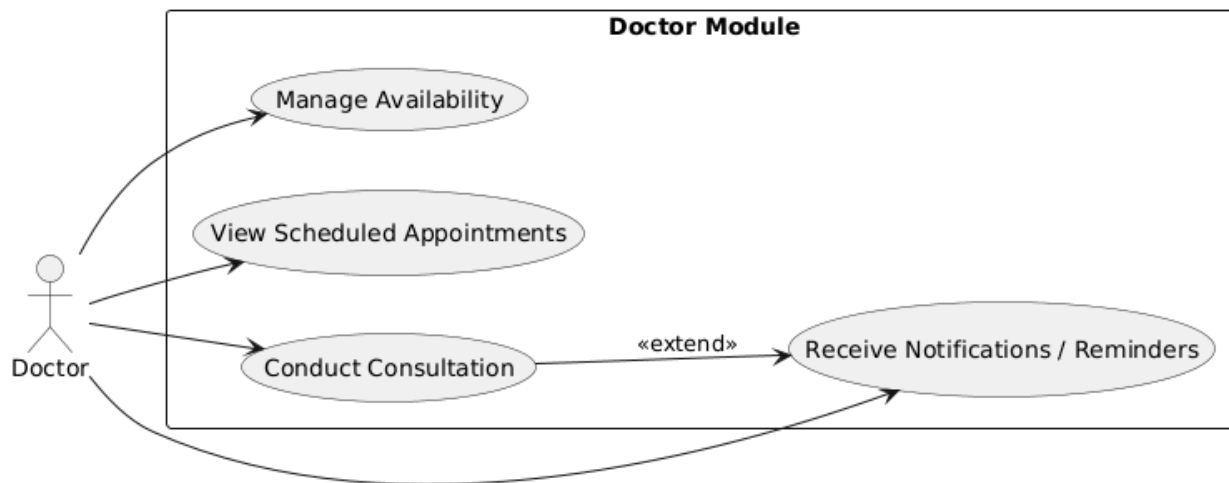


Figure 3 doctor use case diagram

Administrator Use Case Diagram

